

Comm. Skills (SS2010)

Final Exam

Date: May 18th 2026

Course Instructor(s)

Mr. Muhammad Zulfiqar Ali

Total Time (Hrs.): 3

Total Marks: 75

Total Questions: 6

Roll No

EE – 6 A,B,C
Sections

Student Signature

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Attempt all the questions. Do not solve on the question paper. DO NOT change the actual Question Numbers. You may spend about 50 – 55 minutes on each part. Use Bloom's Taxonomy.

PART A: CLO 3: Demonstrate critical reading skills to evaluate the use of statistical tools and conclude findings effectively.

- Q1: Assume that you are a junior engineer at a power distribution station. The following table illustrates the performance of three different Transformer cooling methods over a 24-hour peak load cycle.

Cooling Method	Avg. Temp. Rise (°C)	Efficiency (%)	Maintenance Cost (PKR/month)	Failure Rate (per 10k hours)
ONAN (Natural)	55	92.4	42,300	0.04
ONAF (Forced Air)	42	94.8	90,240	0.02
OFAF (Forced Oil/Air)	30	97.2	239,700	0.01

Construct a formal academic paragraph (150 – 200 words) that evaluates and describes this dataset. Please use the third person, formal, objective tone. Your response must:

- Illustrate the correlation between cooling intensity and system efficiency.
- Calculate and discuss the "cost-to-reliability" understanding and adjustment for a budget-constrained utility service provider.
- Conclude which method is most workable for a high-density urban grid based on the statistical evidence provided. Avoid generic content points.

[15]

Q2: Read the following summary from a pilot study on Renewable Energy Integration for a small-scale microgrid in one of the distal areas of Pakistan:

"The pilot study monitored three power sources over a 12-hour daylight cycle. Solar PV output stayed the most volatile, starting at 5 kW at dawn, peaking at 50 kW at noon, and returning to 10 kW by sunset. In contrast, the Wind Turbine kept a steady, moderate output of 25 kW throughout the day due to consistent coastal breezes. The Battery Storage System (BSS) acted as a buffer; it stayed in 'Charging Mode' during the noon peak (absorbing 15 kW) and switched to 'Discharge Mode' during the evening hours to provide 20 kW to the grid. While Solar PV offered the highest peak capacity, the Wind Turbine provided the highest overall reliability for base-load requirements."

Task: Translate the operational data described in the above written text into a structured Technical Comparison Table or make a System Flowchart. Your answer must:

1. **Tabulate** the output levels (in kW) for all three sources at 3 specific times: Dawn, Noon, and Sunset.
2. **Classify** each energy source based on its "Operational Behavior" (e.g., Volatile, Constant, or Reactive) as described in the passage.
3. **Identify** the specific time-period where the system experiences the highest "Total Grid Contribution" based on the values provided.

[10]

PART B: CLO 4: Identify workplace conflict and resolution strategies.

Q3: Apply conflict resolution frameworks to the engineering scenarios provided below. Your answer should prove professional judgment and a clear understanding of workplace dynamics. Structure your answers logically, using headers or bullet points where needed.

Scenario: You are the Lead Engineer for a substation project. A conflict has appeared between the Design Team and the On-site Implementation Team. The designers insist that a specific protection relay configuration is necessary for safety compliance. However, the site team argues that this configuration causes "nuisance tripping," which is delaying the project deadline and incurring daily liquidated damages. The Project Manager is pressuring you to bypass the safety setting temporarily to meet the project handover date.

Task: Examine the conflicting interests of the three parties involved (Design, Site, and Management). Develop a comprehensive resolution strategy that uses a "Collaborating" (Win-Win) approach. You should:

1. **Illustrate** the potential consequences of choosing one side over the others.
2. **Construct** a solution that meets the deadline without compromising the engineering standards.
3. **Create** a paragraph (80–100 words) for the stakeholders to start the problem solution. [15]

Q4: Case Study: Your engineering department has distributed a limited budget for professional software licenses. Two senior engineers, Ahmed and Zahid, both vital to different high-priority projects, are in a heated dispute. Engineer Ahmed (Power Systems) claims the funds for a transient analysis tool, while Engineer Zahid (Control Systems) demands the funds for a PLC simulation suite. The dispute has escalated to the point where they refuse to share common lab equipment or even work together in similar time slots. The overall situation is now seriously affecting the productivity of the junior engineering staff.

Categorize the type of conflict presented in this scenario (e.g., Task, Relationship, or Value-based). Execute a mediation plan using the Interest-Based Relational (IBR) approach (the win-win situation). Your answer should:

1. Differentiate between their "Positions" (what they want) and the "Interests" (why they need it).
2. Construct three objective criteria to decide the budget allocation fairly.
3. Propose a strategy to restore the collaborative environment in the lab for the junior staff. [10]

PART C: CLO 5: Demonstrate job seeking and interview skills to secure employment opportunities.

Q5: Scenario: During a technical interview for a Smart Grid startup position, the Lead Engineer asks you a "Negative-Behavioral" question:

"Tell us about a time you made a significant technical error during a project. How did you find it, and how did you resolve the situation with your team?"

Construct a structured reply to this question using the STAR (Situation, Task, Action, Result) Method. Your answer should:

1. Identify a realistic engineering-related conflict or error in any assumed/real industrial situation.
2. Calculate the impact the error had on the project's progress or safety.
3. Produce a conclusion that highlights the "Lesson Learned," proving your growth and accountability as a professional engineer. [15]

Q6: Elaborate 'Communication Orientation Motivation (COM)' technique to overcome public speaking anxiety. Use two strong examples to support your answer. [10]